ETHICS IN AI: AN INTERDISCIPLINARY ANALYSIS OF THE SYSTEMIC HARMS PERPETUATED BY AI AND PREDICTIVE POLICING TECHNOLOGIES IN U.S. LAW ENFORCEMENT

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Introduction

Scholars from singular disciplines have been raising significant ethical concerns for marginalized communities in light of the growing integration of artificial intelligence (AI) and related computational technologies in daily operations of socio-political systems like law enforcement (Benjamin 2019; Alexander 2010). However, their critiques, if heeded at all, often trigger technological adaptations that may appear more neutral when, in fact, they further obfuscate the same biased effects. These critiques highlight that harm continues to occur, but without a more robust, interdisciplinary lens, these critiques can miss how, despite supposed social progress, the same harms are perpetuated.

One of the inherent strengths of digital humanities is that the field offers *interdisciplinary* pathways for analyzing these emerging technologies and their impacts on socio-political landscapes. Focusing on predictive algorithms and AI technologies in law enforcement, this paper argues that a digital humanist inquiry into the historical development of law enforcement in the United States is necessary for identifying how emerging policing technologies are the latest adaptation to perpetuate systemic harm against marginalized communities. In doing so, this paper offers a "theory of justification" that tracks the motivation for perceived neutrality or unbiased conduct amidst an entity that continues to disproportionately harm marginalized communities. Specifically drawing from women of color feminist critiques, grounded in intersectional and structural analyses, this paper uncovers how the historical roots and subsequent adaptations in law enforcement theories and methods have maintained systemic inequality by design. Today, these new technologies offer the latest mechanism for adapting to criticism while perpetuating harm.

Methods

This paper an interdisciplinary, digital humanist inquiry that weaves together research from the social sciences and humanities to analyze emerging technologies. Specifically, this paper draws from criminology, theories of criminal justice, feminist jurisprudence, critical race theory, adversarial machine learning, and women of color structural feminisms to understand how emerging technologies fit within a historical pattern of justified systemic harm (Ruíz y Flores 2022; Mills 2025). Other disciplinary contributions that, when brought together, clarify how this pattern has persisted include computer science studies that reveal the non-neutral nature of computer vision (Kalluri et al. 2023). This scholarship notes that mass surveillance in military and carceral contexts has formed the foundational functions and impacts of computer vision, to the express detriment of BIPOC communities.

By bringing this scholarship together, this paper traces the historical development of law enforcement in the United States to situate emerging computational technologies within this history. This paper tracks a gradual shift from law enforcement's formerly explicit role as a mechanism of control over Black, Indigenous, and People of Color (BIPOC) toward servants of the broader public. This shift parallels the changes in theories of crime from criminalizing the allegedly inherent nature of criminals to criminalizing "crime-prone" environments. While these shifts were praised in response to the Civil Rights Era and similar advocacy efforts to enact political change, the effects of targeting and over policing marginalized communities have remained largely unchanged. In fact, New York's Metropolitan Transportation Authority's recent announcement to explore using AI systems in predictive policing to address dangerous behaviors on city subway platforms highlights the ongoing danger that such technologies present to BIPOC communities who, as this paper demonstrates, have consistently been targeted by law enforcement regardless of the tools that police use (Preston 2025). Ultimately, through analyzing the history of criminology theories, legislative efforts, and policy

changes, this paper reveals how emerging technologies in law enforcement are the latest iteration of a longstanding pattern of systemic disenfranchisement of marginalized communities.

Theory of Justification

By situating these technologies in their historical contexts, this paper tracks a "theory of justification" that challenges the perceived neutrality that such technologies present. A theory of justification considers the technology's design and socio-political context to examine the ways in which users rely on such technology to pursue moral absolution from the technology's outputs and decision-making. This theory unmasks alleged "procedural objectivity" in which objective or neutral procedures are thought to produce objective or neutral outputs.

This theory helps explain how institutions like law enforcement can appear to adapt to justice-oriented policy changes while perpetuating systemic harm. For example, law enforcement agencies have introduced racial bias training and protocols, but agencies continue to disproportionately target BIPOC communities (Alexander 2010). A theory of justification explicates why this occurs because the pursuit of moral absolution without changing hierarchies of power and norms prompts adaptations that can help align conduct with new conceptions of justice while maintaining systemic harm through new means, like algorithmic and predictive technologies (Alexander 2010; Tuck and Yang 2012; Benjamin 2019; Mussgnug and Leonelli 2024).

Finally, this theory contributes to discussions around equity, diversity, and inclusivity (EDI) by analyzing the socio-political structures that underpin methods of justification within computational technology. It fosters a critical engagement with digital humanities by examining how these technologies perpetuate inequalities despite attempts to diversify leadership or introduce bias training.

Limitations of AI and Predictive Technologies

By recognizing and tracking this allure of objectivity, this paper reveals how the limitations of AI and predictive technologies are frequently obfuscated. Issues like insufficient or biased datasets, restriction of generalization beyond specific contexts, and the generation of misleading or false information are recognized issues with advanced computational technologies that go unrecognized in contexts such as policing. For example, insufficient or poor-quality data available to train AI models—known as data poverty—can result in biased predictions that disproportionately target these communities, leading to over policing and the exacerbating of social inequalities (Ibrahim et al. 2021).

Likewise, the inherent limitation of AI models, particularly large language models (LLMs), in performing tasks without specific training data – often called lack of zero-shot learning – can lead to inaccurate predictions and disproportionate policing of BIPOC communities. (Tortora 2024; Udandarao et al. 2024). LLMs also pose significant safety risks due to their complexity and broad capabilities that make preventing inaccurate or harmful outputs difficult (Rando et al. 2025). This difficulty stems from the black-box nature of LLMs that makes them incapable of explaining their outputs or predictions. Additionally, the closed-source development methods employed by frontier labs (such as OpenAI, Anthropic, and Google) impede security research and trustworthy safety assessments, which could worsen biases in applications like predictive policing. Moreover, LLMs can "hallucinate" by generating responses that contradict real facts despite receiving accurate prompts (Ziwei Xu, Sanjay Jain, and Mohan S. Kankanhalli 2024). Hallucinations can stem from various issues, such as poor-quality data, biased information, outdated information, and architectural deficiencies with the AI system itself. These issues undermine the reliability of AI systems and exacerbate existing inequalities when applied to policing practices that disproportionately affect marginalized communities.

Law enforcement agencies have a long-standing history of upholding racialized, gendered, ethnic, and class-based targeting. This inherent bias necessarily impacts how predictive technologies operate. Extensive literature documents officer implicit and explicit biases against people of color, those experiencing poverty, immigrants, women, and their intersections (Alexander 2010; Benjamin 2019) However, as this paper demonstrates, these technologies do not adequately address the influence these biases have on its technology's data collection, analysis processes and model outputs.

Conclusion

The criminal justice system continues to evolve and adapt, but, as this paper argues, these adaptations have perpetuated the targeting of marginalized communities while appearing to progress toward neutrality. This "march toward neutrality" is exacerbating the harms that marginalized communities have faced since the earliest forms of the criminal justice system in the United States. Therefore, this paper concludes that no amount of tinkering or technological changes will actually alter the harmful impact of law enforcement's use of them because a theory of justification guides decisions to avoid systemic change. Simply put, the increase in model capabilities is not en-

ough. We also need to ensure that models are safe, free of bias, and lead to equitable outcomes. This will require system-wide change and interdisciplinary collaboration. While efforts to diversify leadership or introduce bias training are beneficial, they must be coupled with a rigorous analysis of technological systems and advocacy toward cultural shifts. Absent this cultural shift, technologies will continue to reflect the longstanding problems embedded in the historical design of law enforcement in the United States. Drawing from these inherent strengths in digital humanities, this paper urges implementing ethical technological practices that promote inclusivity and systemic change.

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